Supplementary material

Spectral bounds and their limits in Pauli contextuality: exact configurations and a four-qubit C3 certificate

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This journal-format document is derived from the complete paper, version 1.2.0. Original section numbers and cross-references are retained; gaps are intentional. The supplement contains: 10. A running example on the Tsotchke stack, 11. Falsifiability and replication, Appendix A. Reproduction, Appendix B. Witness assignments. References are reproduced in both journal documents.

10. A running example on the Tsotchke stack

10.1 State preparation, simulated contexts and quantum geometry

All four components are used for what they are.

- **Moonlab** (v1.2.1, prebuilt WebAssembly engine) is a classical state-vector simulator. It plays the quantum device.
- **quantum_rng** is a classical, "quantum-inspired" integer and trigonometric mixer, not a hardware quantum source. It supplies seeded pseudo-random numbers. Its stream depends on the C library it is built against, so no result depends on its exact values: every count is recomputed independently.
- **QGTL** (quantum_geometric_tensor): only the `diffgeo_*` kernels (Fubini–Study metric, Berry curvature), which we had verified earlier against closed forms, are used.
- **Eshkol** (v1.3.6) is a Scheme dialect with automatic differentiation. It carries an independent implementation of the geometry and the differentiation checks.

Experiment on Moonlab, classical families (`src/moonlab_experiment.mjs`, seed 20261109). A Pauli observable is measured projectively by rotating its eigenbasis onto Z of one qubit (gates H, $S^\dagger$, CNOT), calling Moonlab's single-qubit measurement with a number from quantum_rng, and undoing the rotation.

step	what is done	result
E0	Moonlab's $\langle P \rangle$ against a direct evaluation; repeated measurement	agree to 8.6×10^{-16}; repeats agree

step	what is done	result
E1	ψ_H on 3 simulated qubits; 63 expectation values; signs as an assignment	all $\pm 1/3$ (to 6×10^{-16}); 63 of 315 contexts violated, three through each point
E2	realified ψ_H on 4 simulated qubits	$\pm 1/3$ on 135 real observables, 0 on 120 imaginary ones; 315 of 1575 violated, seven through each point
E3	every context measured sequentially on random states, $N = 2, 3, 4$, three runs each	15/15, 315/315, 5355/5355 products equal to the quantum sign (16065 projective measurements and 161406 gates per four-qubit run)
E4	the same with a random Pauli error of probability p on each qubit before each measurement; 3000 to 4000 sampled contexts	Figure 4 and the table below

The table reports the fraction of sampled contexts whose outcomes multiply to the quantum sign, against proved noncontextual ceilings. For three and four qubits the ceiling is $1 - d_{\text{floor}}/L$. The two-qubit column uses the independently known exact degree $d = 3$, giving $1 - 3/15 = 0.800$; the spectral floor alone rounds to 2 there.

error p	$N = 2$	$N = 3$	$N = 4$
0	1.000	1.000	1.000
0.01	0.979	0.969	0.957
0.02	0.958	0.939	0.930
0.05	0.897	0.867	0.830
0.10	0.805	0.747	0.717
0.20	0.669	0.616	0.591
ceiling, proved	0.800	0.800	0.729
best classical model known	0.800	0.800	0.706

Experiment on Moonlab, the configurations of Table 1 (`src/moonlab_saturated.mjs`, seed 20261110).

step	what is done	result
S1	the first six states of Table 1 loaded as amplitudes; Φ_5 and Φ_7 prepared by running their circuits (Hadamard, controlled-phase and Toffoli gates of Moonlab: 14 and 34 gates); every Pauli expectation value read through Moonlab's gates	the circuits give the phase states to 10^{-16}; non-zero expectation values of one modulus to 10^{-15}; supports and signs equal the masks in `data/flat/`; contexts and violated contexts as in Table 1, including 2 824 192 and 729 344 for seven qubits
S2	every context of S_B (three runs), of S_5 and of the support of Φ_5 measured sequentially on random states; for S_7 a sample of 30000 contexts, and once all of them	1095/1095, 10880/10880, 10880/10880, 30000/30000 products equal to the quantum sign; in the full seven-qubit run 2 824 192/2 824 192 (8 472 576 projective measurements, 161 550 336 gates)
S3	the same with Pauli noise; 4000 sampled contexts	the table below

Because the degrees are exact, the ceilings in this table are exact as well: $1 - d/L$. For the four- and five-qubit configurations they are much higher than for the full space, so the test tolerates

much less noise; for S_7 the ceiling is 0.742 and the simulated test stays above it up to an error probability of two per cent per qubit and measurement.

error p	S_B (four qubits)	S_5 (five qubits)	support of Φ_5	S_7 (seven qubits)
0	1.000	1.000	1.000	1.000
0.002	0.992	0.989	0.990	0.986
0.005	0.982	0.981	0.978	0.963
0.01	0.962	0.955	0.949	0.938
0.02	0.929	0.913	0.916	0.877
0.05	0.835	0.807	0.797	0.733
0.10	0.719	0.679	0.670	0.597
ceiling, exact	0.932	0.956	0.956	0.742

These are frequencies in a seeded software simulation. They illustrate the inequalities and are not measurements on hardware or a loophole analysis.

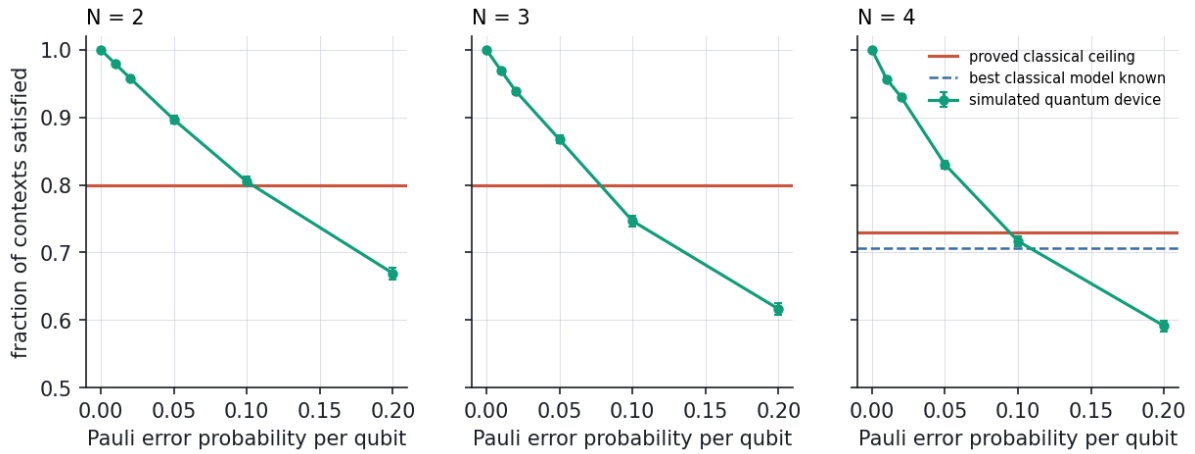


Figure 4. Simulated success fraction on Moonlab against the classical ceilings, for the full space.

QGTL (`src/qgt_hoggar.c`, `src/qgt_flat.c`). The kernels `diffgeo_compute_fubini_study` and `diffgeo_compute_berry_curvature`, applied to ψ_H and the 63 tangent vectors $-\frac{i}{2}P_v\psi_H$, return the values of Proposition 3 to 10^{-16}, and the 189 pairs with $|g_{uv}| > 1/12$ are exactly the pairs on the 63 violated contexts. For the eight states of Table 1 and the tangent vectors of their supports they return the three values of Proposition 8 to 10^{-16}; the pairs above the threshold recover the 75 violated contexts of S_B (225 pairs of 9045), the 480 of S_5 and of the support of Φ_5 (1440 pairs of 122760), and the 729 344 of S_7 (2 188 032 pairs of 33 028 128), and no others.

Eshkol (`src/eshkol/hoggar.esk`, `src/eshkol/saturated.esk`). An exhaustive pass over all 2^{15} two-qubit assignments gives minimum 3, attained 640 times. The Hoggar signs violate 63 contexts. Reverse-mode automatic differentiation of the cubic form at the Hoggar Pauli vector x^* gives $T(x^*) = 7 = (D-1)(D-2)/6$ and $\nabla T(x^*) = 3x^*$ to 7×10^{-16} (the Lagrange condition for a maximizer on the sphere); nested forward-mode second derivatives along four tangent directions are negative (-0.59 to -0.74). For the pivot state the second script finds 135 non-zero expectation values, 1095 contexts and 75 violated ones, $T = 35 = (D-1)(D-2)/6$, and $\nabla T = 7x$ to 1.3×10^{-15}, which is condition (d) of Theorem 4.

10.2 Exact C3 arithmetic in Eshkol

The separate serial compiled run uses Eshkol 1.3.6-evolve through `eshkol-run -r`. It checks $A_0M = MA_1$ and both $MR = RM = 46656I$, then evaluates the block-moment formula to

obtain 14976 and 15360. An arbitrary-precision Python replay bounds every input, product and ordered accumulation of this program by 408304 in absolute value, below $2^{63} - 1$. The recorded run returned zero with empty standard error; `results/frontier-c3-eshkol.json` records its command, hashes and process-local paths to the existing LLVM 21 and CUDA 12.9 installations. This qualifies the tested serial arithmetic execution, not compiler correctness or GPU/parallel behavior.

11. Falsifiability and replication

11.1 Floor, saturation and simulated measurement checks

Predictions that could fail. (P1) No assignment violates fewer contexts than the floor. (P2) The saturated cases are reached exactly. (P3) For every S and f, $\operatorname{tr} A = 0$, $\operatorname{tr} A^2 = D|S|$, $\operatorname{tr} A^3 = 6D\beta$. (P4) On an elliptic quadric the spectrum of A is doubly degenerate. (P5) For every pure state $T(p) = (D-1)(D-2)/6$. (P6) Ideal sequential measurements reproduce every context sign in every state. (P7) For a flat state, condition (d) of Theorem 4 holds. A counterexample to P1 would provide a finite assignment whose violation count can be recomputed independently from the context definitions.

Tests (`scripts/run_all.sh`).

- `hoggar exhaust2`: all 32768 two-qubit assignments; minimum 3 (640 assignments); for the square, minimum 1.
- `scripts/run_falsify.sh`: 20 runs over $N = 2, \ldots, 5$ and the three families; 20000, 5000 and 500 uniformly random assignments and 1500, 200 and 12 annealed ones per configuration for $N = 3, 4, 5$. No assignment below the floor. Minima reached by annealing: 63, 21, 9 ($N = 3$); 1575, 315, 315 ($N = 4$); 31479, 6975, 7087 ($N = 5$). These equal the best known values in every case.
- A negative control: the false claim $d(W(5, 2)) \geq 64$ is refuted by the Hoggar witness with exit code 3, showing that the harness can fail.
- `scripts/flat_states.py` (68 checks, integer arithmetic in $\mathbb{Z}[i]$ except one labelled check): for the six states of Table 1, the single modulus, the support, the number of contexts, the number violated, the floor, condition (d) at every point and the vanishing of every outside sum; the statements of Theorem 5 about S_B; Proposition 6 on all 4096 phase patterns; Proposition 9 for one to four qubits; and an annealing search from random starts on S_B and S_5, which reaches 75 and 480 and never goes below.
- `build/third_level` (seven seconds): the enumeration of Theorem 12, all 33 554 431 forms for five qubits included, with the orbit computation; it fails if any count differs from the table.
- `scripts/phase_states.py` (81 checks, integer arithmetic): for $CS|{+}{+}\rangle$, $CCZ|{+}{+}{+}\rangle$, $CS\,CS|{+}{+}{+}\rangle$, Φ_5 and Φ_7, the constant rank, the single modulus, the support, the numbers of contexts and violated contexts, condition (d) and the outside sums, and the description of the support in Proposition 11; Proposition 10 for all eleven states of this paper; the enumeration of Theorem 12 for $N \leq 4$ by a second implementation; the two symplectic maps of Theorem 12; and the finite-field descriptions of the five- and seven-qubit forms.
- `scripts/run_family7.sh` (about eight core-hours, not part of the everyday run): the enumeration of Theorem 13, the orbit computation, and the random census of the other types; its output is `results/family7.log`. The two counts agree: 13 271 040 solutions found, and $92\,897\,280/7 = 13\,271\,040$ for one orbit with a stabilizer of order 7.
- The seven-qubit numbers 8128, 2 824 192 and 729 344 are computed three times without shared code: in Python with exact integers, in C from the masks (inside the QGTL program) and in JavaScript from the expectation values returned by Moonlab.

- `scripts/analysis.py` (62 checks) and `scripts/ladder.py` (14 checks): P3 on random data to machine precision; P4 on all 120 four-qubit elliptic quadrics; P5; the geometry of the violated sets; closed-form counts against enumeration; agreement of the C and Python integer bounds.
- Witness files (`data/witness/`, `data/flat/`) are recounted by independent routines that use no tables.

Independent implementations. The geometry, the context signs and the counts are implemented five times: C (`src/hoggar.c`), Python with explicit matrices or exact integers (`scripts/analysis.py`, `scripts/flat_states.py`), JavaScript (`src/moonlab_experiment.mjs`, `src/moonlab_saturated.mjs`), Eshkol, and the Lean model, in which the signs are theorems about matrix products. Their overlapping checks agree on the counts recorded for those tests. This does not mean that each implementation checks every result: in particular, the seven-qubit context counts are outside the kernel evaluation, and the C3 checks use the separate programs described next.

11.2 Independent C3 reconstruction and negative controls

The literal-matrix verifier builds each four-qubit Pauli from the 2×2 matrices, selects the real quadric and section, and computes exact carrier characteristic polynomials. A change-of-basis matrix T with $T^{\mathsf{T}}T = TT^{\mathsf{T}} = 8I$ intertwines the completion matrix and the block Jordan representation, checking that the coordinate transformation loses no component.

A separate verifier constructs the completion matrix from all 1120 outside signed contexts. It checks commutation and multiplies each triple to determine its sign, then recomputes both carrier polynomials and completion cube traces. This route does not use (C7). Its last three tensor factors are reversed relative to the printed convention; that fixed orthogonal basis permutation is recorded explicitly and preserves the represented assignments and spectra. Pauli entries $0, \pm 1, \pm i$ are exact in their initial representation, real matrices are converted to integers, and product bounds fit signed 64-bit arithmetic. Characteristic polynomials use exact SymPy arithmetic, not eigenvalue tolerances.

Flipping one off-diagonal context entry and its symmetric partner changes the two cubic traces to 14808 and 15144 respectively, so the expected certificate fails. The verifier rejects Python `-O`, which would disable its assertions. These are specified negative controls, not a proof that every possible implementation error is detected.

The interactive C3 model independently constructs context signs, computes characteristic polynomials with a BigInt Faddeev–LeVerrier recurrence and evaluates cube traces with BigInt arithmetic. Editing a sign triggers a fresh computation. A modified assignment is not required to retain cospectrality; the original four-negative-word witnesses are the exact claim being tested. The experiments concern finite mathematical operators and are separate from the Moonlab measurement simulations.

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Appendix A. Reproduction

```
# engine, random source, QGT programs (gcc; no other dependency)
bash scripts/build.sh
# every test: exhaustive N=2, falsification suite, flat states, analysis, ladder, Moonlab, QGTL,
↪  Eshkol
bash scripts/run_all.sh
# Lean (needs elan; fetches Lean 4.35.0 and the Mathlib cache)
cd lean && lake exe cache get && lake build
# the exact degrees of Section 5 (kernel evaluations; see the CI evidence for time and memory)
lake build HoggarBound.Saturated HoggarBound.SaturatedPhaseFive
lake env lean SaturatedVerification.lean
# the seven-qubit enumeration of Theorem 13 (about eight core-hours)
cd .. && bash scripts/run_family7.sh
```

The C3 verification programs require Python, NumPy, SymPy and Node.js. From the repository root:

```
python scripts/frontier_c3_verify.py
python scripts/frontier_c3_context_verify.py
node tests/frontier_c3_browser_test.cjs
python scripts/frontier_c3_integer_bounds.py
python scripts/unified_section_score.py
python scripts/frontier_c3_demo.py
```

The last command requires Eshkol; set `ESHKOL_RUN` when it is not on the system path. The wrapper generates `build/frontier_c3_demo.esk` and invokes `eshkol-run -r`, retaining its process-local settings in the result record. The published similarity certificate, not a random search, is the input to these arithmetic checks.

From `lean/`, build the C3 targets sequentially and then audit the declarations:

```
lake build HoggarBound.C3Data
lake build HoggarBound.C3Certificate
lake build HoggarBound.C3Geometry
lake build HoggarBound.C3Theorem
lake build HoggarBound.C3Spectra
lake env lean C3Verification.lean
```

The recorded Windows finite-certificate check took approximately 61 minutes. This is proof-checking cost, not the running time of the interactive demonstration or a physical quantum process. The direct context check used Python 3.9.13, NumPy 2.0.2 and SymPy 1.14.0. Raw-byte and LF-normalized input hashes distinguish local Windows line endings from Git-normalized text.

Appendix B. Witness assignments

`data/witness/nN_t.txt` holds, for N qubits and family t (`f` full, `h` hyperbolic, `e` elliptic), one header line and a hexadecimal mask whose bit v is set when $f(v) = -1$, with $v = a + 2^N c$. The three-qubit Hoggar witness produced by the Moonlab run is

```
3 f 63
163e5efe76fefe00
```

`build/hoggar verify N t file` recounts any witness from the definitions.

`data/flat/nN_name.txt` holds the configurations of Table 1: a header line with N, the numbers of observables and contexts and the degree; the mask of the configuration; and the mask of the optimal assignment. For the pivot configuration:

```
4 135 1095 75
b4451eef7889d22344b5ee1f887922d34b45e1ef87892d23bbb5111f7779ddd2
20011cc01888c2204424221f001100c34244212f01010c038880111177790000
```

The same two numbers are `pivotMask` and `pivotNeg` in `lean/HoggarBound/FlatData.lean`. The files `n5_phase5.txt` and `n7_phase7.txt` hold the supports of Φ_5 and Φ_7 in the same format; the seven-qubit masks have 4096 hexadecimal digits.

For C3, the four negative labels of each carrier are given in Table 3. The principal machine-readable records are `results/frontier-c3-n4-pair.json`, `results/frontier-c3-n4-verification.json`, `results/frontier-c3-n4-completions.json` and `results/frontier-c3-context-verification.json`. They record the matrices, counts, exact polynomials and moments, mutation controls and dependency versions. The theorem's assignments are fully defined in Section 8.2 independently of those files.